Building 6G ISAC in Practice: From Raw Radio Signals to AI-Driven Sensing

From Connectivity to Awareness from physical environment

Mobile networks have traditionally been designed to connect people, devices, and services. Each generation has improved data rates, latency, reliability, coverage, and capacity. In 6G, however, the role of the network is expected to expand further. A 6G network may not only deliver information, but also help understand the physical environment in which communication takes place.

Integrated Sensing and Communication, or ISAC, is one of the key technologies behind this shift. The basic idea is simple: radio signals used for communication can also carry information about the surrounding environment. By analyzing reflections, propagation characteristics, and measurement results, the network may detect objects, track movement, or understand changes in the radio environment.

The important point is that ISAC is not just about adding radar-like functions to a base station. Its broader value lies in reusing communication infrastructure, radio resources, and signal processing capability to create a distributed sensing layer. If designed carefully, 6G ISAC can expand the value of mobile networks beyond connectivity without requiring a completely separate nationwide sensing infrastructure.

This is why ISAC is becoming an important topic in 6G. It connects several long-term directions: network intelligence, service expansion, coverage-aware operation, energy and deployment efficiency, and tighter integration between the physical and digital worlds.

What ISAC Means for 6G Networks

The value of ISAC starts from infrastructure reuse. Mobile networks already have wide-area deployment, radio coverage, synchronization, beamforming capability, and baseband processing resources. If sensing can be integrated into this existing communication framework, the network can provide sensing capability with a lower incremental deployment burden than building a separate sensing system from scratch.

From this perspective, ISAC can be understood as a new value layer on top of the communication network. A base station, a fixed wireless access device, a vehicle, or another network-connected node may contribute to sensing, depending on its capability and deployment condition. The sensing result may then be used by applications, by the network, or by other devices.

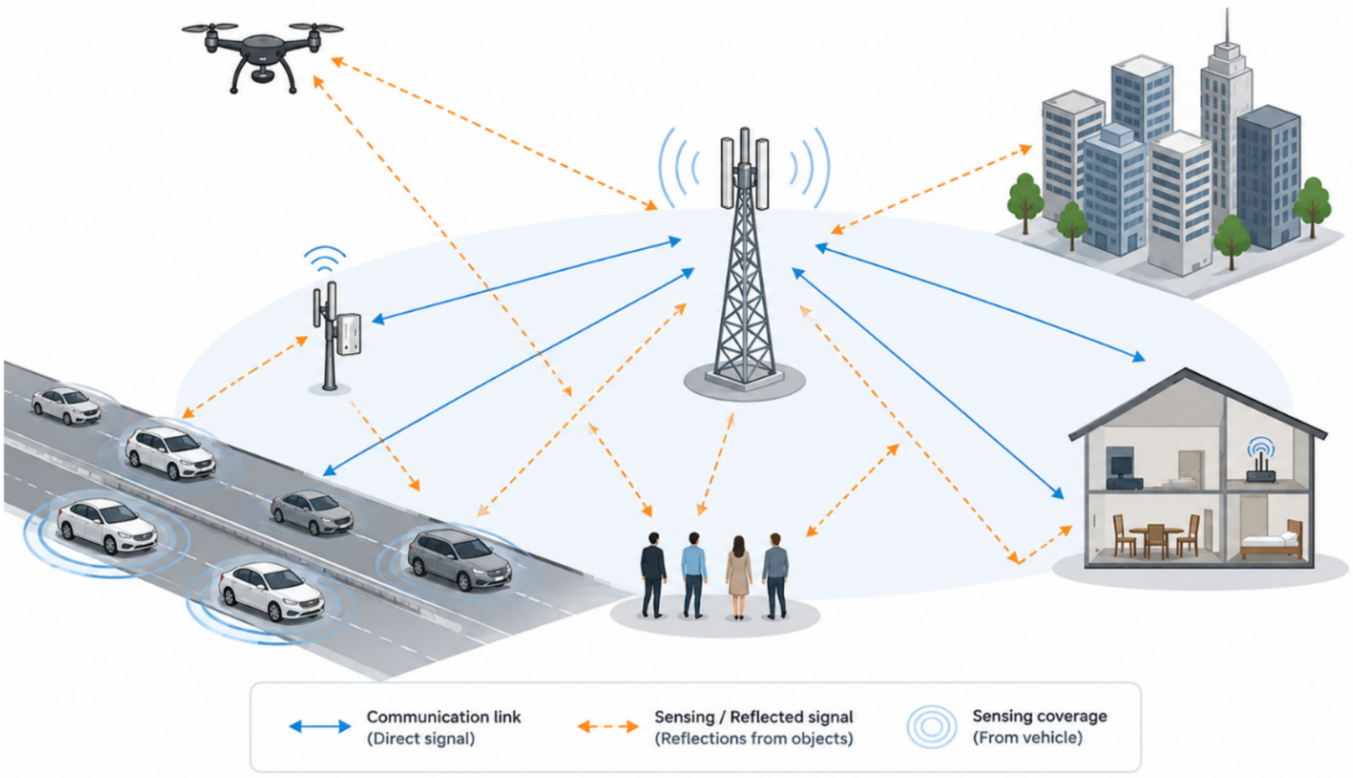
This also creates new business opportunities for 6G. Today, the main value of mobile networks is still largely tied to connectivity: data plans, broadband access, enterprise private networks, and device connections. ISAC can expand this value by enabling sensing-based services on top of the same infrastructure. For example, environmental awareness, road and traffic monitoring, industrial site monitoring, public safety support, indoor presence detection, and context-aware network operation may become new service opportunities if sensing information can be provided in a reliable and privacy-aware manner.

The business value of ISAC does not come from replacing dedicated sensors or radar systems in every scenario. Instead, it comes from enabling “good-enough and widely available” sensing capability where communication infrastructure already exists. This distinction is important. A dedicated sensing system may provide higher accuracy for a specific mission, but it often requires separate deployment, operation, and maintenance. ISAC can create value by offering scalable sensing coverage with lower additional deployment effort, especially for use cases that benefit from wide-area observation rather than extremely high sensing precision.

This can be particularly meaningful for operators and ecosystem partners. For operators, ISAC may provide a path to move beyond pure connectivity revenue and offer sensing-enabled services to enterprises, municipalities, mobility platforms, home service providers, and industrial customers. For device and infrastructure vendors, ISAC may create new requirements for sensing-capable radio design, reporting frameworks, edge processing, and service platforms. For application providers, it may open a new source of environmental and contextual information, provided that privacy, security, and data ownership are properly addressed.

This does not mean that every communication node should become a high-performance radar. A practical 6G ISAC design needs to respect the constraints of communication systems. Sensing should be implementable, as much as possible, using common RF and baseband modules. This principle is important because the real strength of ISAC is scale: useful sensing capability should be added without making the communication network significantly more complex or costly.

A dedicated radar system may provide strong sensing performance in a specific location. A communication network, on the other hand, can provide broad and continuous coverage if sensing is integrated efficiently. The design goal should therefore be to find a practical balance: useful sensing capability, reasonable system impact, acceptable cost for communication operation, and clear service value for the 6G ecosystem.



From 5G ISAC to 6G ISAC: Why the Scope Needs to Expand

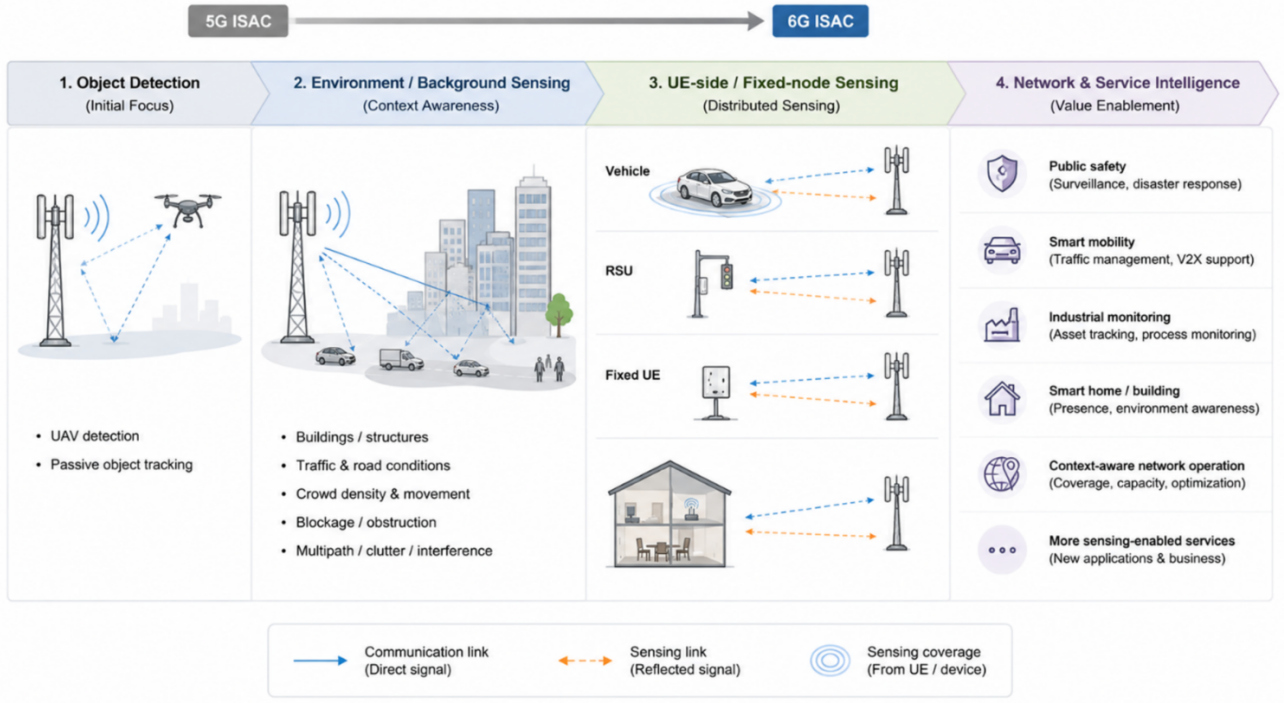
In 5G-related ISAC studies, UAV detection has often been used as an intuitive and representative example. This makes sense. UAV detection is easy to understand, and it clearly shows how a communication network may be used to detect or track passive objects without deploying dedicated radar infrastructure.

UAV detection remains an important use case for 6G ISAC. However, if ISAC is discussed only through UAV detection, its value may look too narrow. The broader opportunity for 6G is environmental awareness. A 6G network should be able to understand not only whether a specific flying object exists, but also how the surrounding environment affects communication and services.

This includes buildings, roads, vehicles, crowds, blockage, reflection, clutter, and other environmental factors. A vehicle passing through a street, a crowd forming in a public space, a moving blockage in an indoor environment, or a change in surrounding reflectors may all affect radio links. ISAC can provide an additional source of information about these changes.

In other words, 6G ISAC should move beyond object detection alone. It should become a framework for sensing the environment around communication networks.

Key Use Cases for 6G ISAC



1. Environment and Background Sensing

A practical starting point for 6G ISAC is environment or background sensing. Instead of focusing only on a specific target, the network may collect information about the surrounding radio environment. This can include static objects such as buildings or walls, non-static objects such as vehicles or people, and environmental factors such as blockage, multipath, clutter, and interference.

This type of sensing can support both services and network operation. For example, background sensing may help identify traffic congestion, crowd density, or changes in the local environment. It may also help the network understand why a link is degraded: whether the cause is blockage, mobility, interference, or a change in the propagation environment.

From a communication-assistance perspective, this type of environmental information can be useful because it is directly related to channel behavior. Environmental sensing may provide information on multipath, blockage, fading, clutter, and interference. Such information may potentially help communication-related functions, but the benefit should not be overstated.

Sensing information does not automatically improve communication performance. How the network uses such information for scheduling, beam management, mobility, or link adaptation may depend on implementation and system design. For the early stage of 6G ISAC, a more realistic goal is to define what sensing information can be measured, how it can be reported, and what level of reliability can be expected.

This is why background sensing is important. It can become the foundation for many future ISAC services without forcing the system to assume one specific application from the beginning.

2. Vehicle, RSU, and Fixed-UE Sensing

UE-side sensing is another important direction for 6G ISAC. However, not all UEs are equally suitable for sensing. A handheld smartphone has strict constraints: limited antenna size, limited transmit power, battery constraints, mobility, and uncertain position or orientation. These factors can make sensing information less reliable, especially when accurate location or angle information is required.

This is why 6G ISAC should look beyond smartphones. Vehicles, road-side units, customer premises equipment, and other fixed or semi-fixed devices may be more attractive sensing nodes.

A vehicle, for example, has several advantages over a handheld device. It can support better antenna placement, has a more stable power supply, and can continuously observe its surroundings while moving through roads. Vehicle-mounted sensing may help monitor nearby objects, road congestion, surrounding buildings, local blockage, and traffic density. This information may be useful not only for the vehicle itself, but also for broader network or service-level awareness.

Road-side units and fixed UEs can provide even more stable sensing information. Their location and orientation can be known or controlled, which improves the confidence of sensing results. This distinction is critical for 6G ISAC. “UE sensing” should not be treated as a single category. A smartphone, a vehicle, an RSU, and a fixed indoor device have different antenna capabilities, power budgets, processing assumptions, mobility patterns, and reporting feasibility. A practical 6G ISAC framework should define these assumptions clearly.

3. Indoor CPE-Based Sensing

Indoor fixed devices, such as CPEs used for fixed wireless access, may also become meaningful sensing nodes in 6G. Unlike handheld devices, CPEs are usually installed in a fixed location, connected to a stable power source, and may have a larger form factor. These characteristics make them more suitable for continuous sensing than battery-limited mobile devices.

A fixed CPE can act as a stable observation point in an indoor environment. Potential home-oriented use cases may include presence detection, indoor environment awareness, safety monitoring, or detection of unusual movement patterns. These examples should be treated carefully, because indoor sensing is closely related to privacy.

A useful ISAC system should not simply collect as much information as possible. It should define what is measured, what is reported, where the sensing function is processed, and how sensitive information is protected.

For this reason, CPE-based sensing may become a good example of the balance required in 6G ISAC. The device characteristics are favorable, the use cases are understandable, and the deployment model is practical. At the same time, privacy, reporting overhead, and processing responsibility must be considered from the beginning.

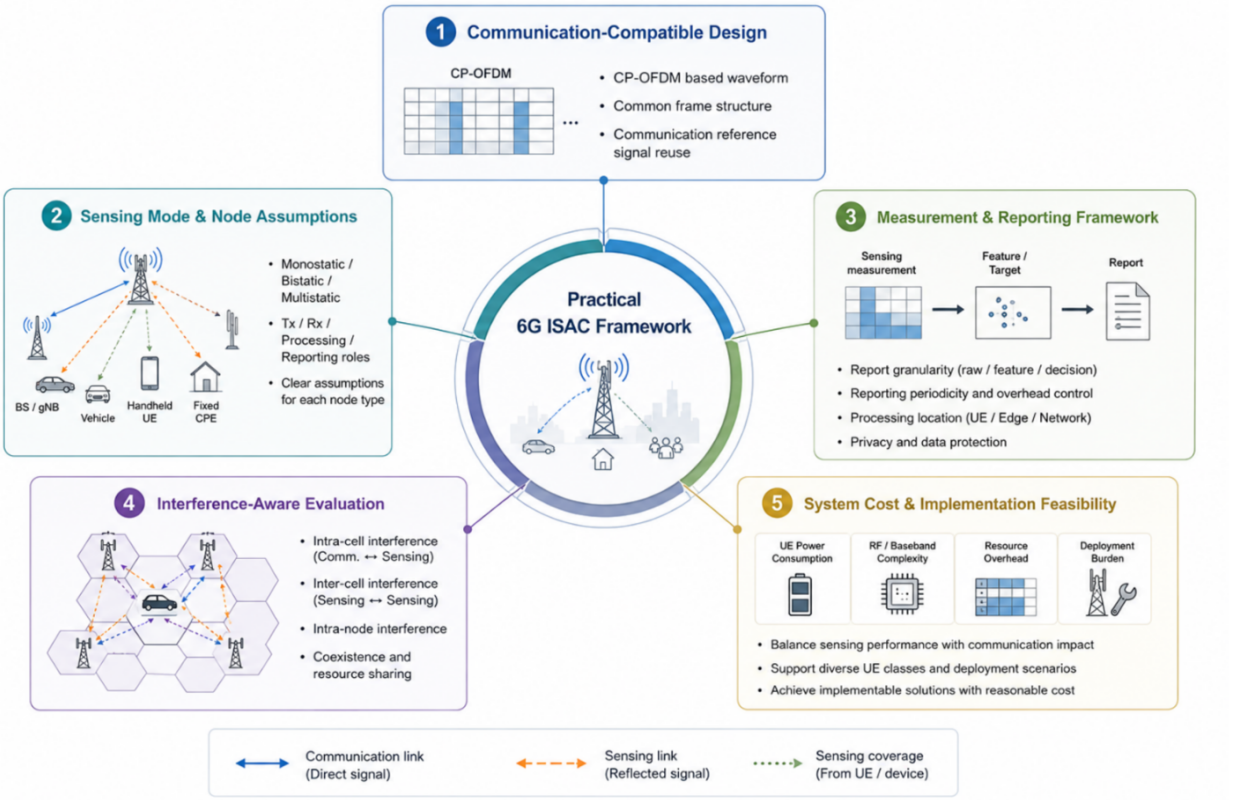
4. Communication Assistance

Communication assistance is another important use case for ISAC, but it requires careful framing. The intuitive idea is that if the network understands the environment better, it may operate communication links more efficiently. For example, sensing information may help identify blockage, moving objects, reflectors, or changes in the propagation environment.

This information could potentially support beam management, CSI acquisition, mobility handling, or interference management. However, it is not yet sufficient to assume that sensing will directly guarantee communication performance improvement in all scenarios.
A practical approach is to separate two questions. The first question is: what sensing information can be measured and reported reliably? The second question is: how should the network use that information to improve communication? In the early stage of 6G ISAC, the first question should come before the second.

This approach avoids overloading ISAC with too many communication-performance assumptions, while still allowing sensing information to become useful for future network optimization.

What 6G ISAC Needs to Address



For ISAC to become a practical 6G feature, 6G ISAC needs to focus on integration with communication from day one. The key question is not how to design the best standalone sensing waveform in isolation. The key question is how to support useful sensing capability within a communication system that must also serve users, manage interference, control overhead, and remain implementable.

Start from Communication-Compatible Design

6G ISAC should start from the communication framework. CP-OFDM-based waveform generation, common frame structure, and communication reference signal reuse should be evaluated before introducing sensing-specific designs. For sensing RS based operation, the signal characteristics may be treated in a similar manner to communication RSs, since they are mainly determined by sequence design and resource mapping.

This does not mean that no enhancement is possible. If a clear sensing performance gap is identified, enhancements to reference signal design and overall procedures, including sequence design, resource mapping patterns, repetition structure for possible range extension, resource configuration, or measurement reporting procedures may be considered. For example, ZC-based sequence design or frequency domain mapping may be considered within the CP-OFDM framework if existing communication RS are not sufficient for sensing. But these enhancements should be justified by realistic evaluation and should preserve communication compatibility as much as possible.

The reason is straightforward. ISAC is valuable because it is integrated with communication. If sensing requires a separate waveform, separate frame structure, or separate hardware assumption from the beginning, the deployment benefit becomes much weaker. A communication-compatible baseline keeps the system scalable and implementable. With this baseline, necessary enhancements may be achieved by optimizing the design of communication signals within the communication framework. Such an approach can allow the network to provide meaningful sensing capabilities without requiring hardware modifications to the communication transceiver.

Clarify Sensing Modes and Node Assumptions

6G ISAC also needs clear assumptions on sensing modes and node roles. Monostatic, bistatic, and multistatic sensing have different technical implications. The transmitter, receiver, sensing function, and reporting node may be the same or different. Consequently, requirements for synchronization, beam operation, interference, and reporting paths also vary depending on the sensing mode.

For BS-centric sensing, standard constraints, such as the need for standardized waveform, should be carefully assessed. For BS monostatic sensing, introducing specification constraints can prevent vendor-specific optimizations needed to handle interference across different deployment scenarios. For BS-BS bistatic sensing, the requirement for stringent synchronization presents practical challenges when implemented across different infrastructure vendors via inter-vendor collaboration. Therefore, relying on vendor implementation might be a better approach for those sensing modes than imposing standard constrains.

On the other hand, for UE-involved sensing, the assumptions become even more important. Various UE types, such as handheld UE, vehicle, fixed CPE, and RSU, should not be modeled in the same way. Their antenna capability, transmit power, processing capability, mobility, and reporting feasibility are different. Furthermore, when both BS and UE are involved (e.g., BS-UE or UE-BS bistatic sensing), the impact on UE operation and the integration with the communication framework become more significant.

Therefore, before defining detailed signal designs, 6G ISAC needs to clarify who transmits, who receives, who processes, and who reports sensing information. Understanding these limitations and device variations can help 6G standardization focus on essential standard impacts, while leaving deployment flexibility to implementation.

Build a Measurement and Reporting Framework

6G ISAC also needs a practical measurement and reporting framework. Sensing results can have very different forms. They may be raw samples, delay-Doppler-angle profiles, detected peaks, or processed target-level metrics. More detailed information may improve sensing quality, but it also increases reporting overhead, processing burden, and privacy risk.

This is especially important for UE-side sensing. If a UE needs to report large sensing data frequently, the overhead may become too high. If the UE processes too much information locally, device complexity and power consumption may increase. Depending on the reporting form, additional post-processing may be required at the aNB side. In such cases, additional contextual information such as sensitive location or environment information, may also be requested to interpret the sensing results report. If such information is included in the report, privacy concerns become significant.

The right design point may differ by use case. A vehicle monitoring road congestion may need frequent but compact reports. A fixed indoor CPE may require stronger privacy protection and local processing. A TRP-side sensing use case may allow more network-side processing but may require stronger coordination across cells. 6G ISAC needs to define flexible but manageable reporting mechanisms that reflect these differences.

Evaluate Interference Realistically

6G ISAC needs realistic interference modeling from the beginning. Sensing signals do not exist in an empty environment. They coexist with communication traffic, neighboring cells, other sensing operations, and reflections from background objects.

This is particularly important for multi-node or cooperative sensing. For example, when BS monostatic sensing is performed simultaneously in different cells, a sensing signal transmitted by one BS may act as interference to echo reception at another BS. Similarly, different cells may operate the same or different sensing modes. In UE-involved sensing, UE transmission for sensing may also create new interference and coexistence conditions depending on how sensing reference signals are transmitted, received and reported.

Without interference-aware evaluation, sensing performance may look promising in simulation but become fragile in real deployment. 6G ISAC therefore needs to evaluate not only ideal sensing accuracy, but also sensing reliability under realistic intra-cell, inter-cell, and intra-node interference conditions.

Balance Sensing Gain and System Cost

Finally, 6G ISAC needs to balance sensing gain against communication impact and implementation cost. A sensing-specific design may improve a particular sensing metric, but it may also increase resource overhead, reduce scheduling flexibility, require additional RF capability, or increase UE power consumption.

This trade-off is especially important because ISAC is intended to be integrated with communication. A design that works well for sensing alone may not be suitable if it consumes too many communication resources or creates too much device complexity.

Therefore, 6G ISAC should evaluate candidate designs across three dimensions at the same time: sensing performance, communication impact, and implementation feasibility. Only when all three are considered together can ISAC become practical for commercial networks.

Conclusion

ISAC is one of the technologies that can expand the role of 6G networks beyond connectivity. Its value is not limited to detecting a specific object such as a UAV. The broader opportunity is to make mobile networks more aware of the physical environment around them.

For this reason, 6G ISAC should focus on practical and scalable use cases: environment and background sensing, vehicle and fixed-UE sensing, indoor CPE-based sensing, and carefully scoped communication assistance. These use cases can build on the existing strength of mobile networks: broad coverage, distributed infrastructure, radio resources, and signal processing capability.

At the same time, ISAC must be designed with realistic constraints. Sensing should be integrated with communication, not isolated from it. CP-OFDM-based waveform generation, common frame structure, communication reference signal reuse, clear UE assumptions, compact reporting, privacy awareness, and interference-aware evaluation should be considered from the beginning.

The key message is simple: 6G ISAC should not be about turning every base station into a standalone radar. It should be about making the communication network aware of its environment in a practical, scalable, and deployable way.

Continuation:

In modern mobile networks, the base station (BS) continuously estimates the radio channel to each user equipment (UE) to support operations such as beamforming, link adaptation, and scheduling. This channel state information (CSI) carries more than link quality: walls, doors, and moving people all shape the paths a signal takes, so changes in the CSI over time reveal what is happening in the surroundings. Integrated sensing and communication (ISAC) exploits this to let the network sense its environment with the same signals and hardware it uses for communication [2]-[4]. We consider the UE-to-BS uplink bistatic configuration, in which the BS receives uplink reference signals from a UE and uses the resulting CSI both for communication and for sensing people at a building entrance.

ISAC is being studied in 3GPP for both 5G-Advanced and 6G [5], [7]. Because cellular networks already provide wide-area coverage, wideband radios, and massive multiple-input multiple-output (MIMO) arrays, sensing added in software can become available wherever the network is, without cameras or dedicated radar hardware. The uplink bistatic mode is particularly attractive for early deployment, since it requires neither full-duplex operation nor new hardware. In this work, we target presence and entry/exit detection on a commercial virtualized radio access network (vRAN): a fixed UE placed indoors at a building entrance transmits uplink reference signals to a commercial 64-transmit/receive (64TR) massive MIMO base station across an outdoor street, and the system detects in real time whether anyone is present near the entrance and whether people are entering or exiting.

Moving from concept to a working system, however, raises challenges at two levels. At the signal level, CSI produced for communication is not radar data. The UE is synchronized to the BS only within communication tolerances, and its residual timing and phase offsets, harmless for communication, appear as false range and Doppler variation when CSI is compared across symbols; together with sidelobes and strong static clutter, they bury weak human reflections. The resulting range-Doppler maps (RDMaps) also differ from natural images: their axes have physical meaning, multipath creates ghost copies of targets, and moving structures such as doors produce Doppler of their own. Apart from an empty scene, a person entering, a person leaving, and someone present nearby are difficult to tell apart in a single RDMap.

At the system level, full-band CSI amounts to hundreds of Mbit/s per UE and must be moved and processed in real time without burdening the radio access network (RAN). Moreover, frame-level outputs fluctuate, yet the reported state must remain stable while still following people as they pass. The remainder of this post introduces the system we built to address these challenges and validates it through experiments at two campus entrances, real-time operation on a commercial vRAN, and demonstrations to several mobile network operators.

2. Proposed System

2.1 System Overview

Figure 1 illustrates the receiver processing chain of our UE-to-BS bistatic ISAC system. The uplink signal transmitted by the UE passes through the receive (RX) front-end and channel estimation stages of the standard communication receiver. At the output of channel estimation, the processing splits into two paths. The upper path is the conventional communication path, where the CSI is used for operations such as scheduling and beamforming. The lower path, highlighted as the sensing block, reuses the very same CSI: a signal preprocessing stage converts it into a clean range-Doppler map (RDMap), and an AI-based event detection stage classifies the RDMap into target events such as a person entering or leaving, or someone being present nearby.

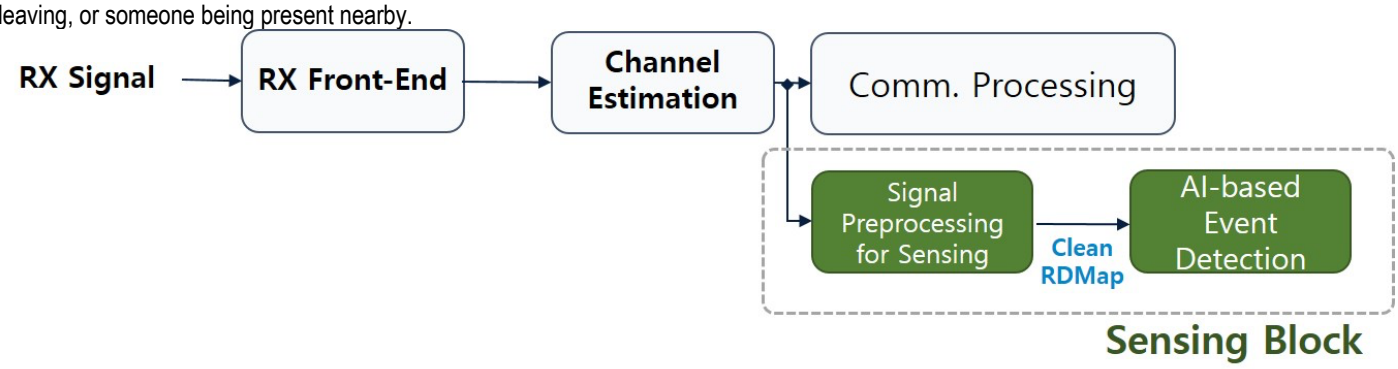


Figure 1. Overall receiver processing chain of the UE-to-BS bistatic ISAC system.

The key design principle is that sensing functionality is added to an existing communication receiver without any dedicated radar hardware or modification to the communication path. The communication processing itself is unaffected; the sensing block simply taps into what the receiver already produces, using uplink reference signal resources for sensing.

In the UE-to-BS bistatic configuration, the base station receives the uplink signal transmitted by a UE. This signal contains both the line-of-sight (LoS) (or otherwise dominant) component, which travels directly from the UE to the BS, and components reflected from surrounding objects such as people. From the RX front-end onward, communication and sensing share exactly the same hardware and the same signal.

Using the uplink reference signal, the receiver estimates the frequency-domain channel response, i.e., the CSI. From the communication perspective, this CSI supports uplink scheduling and link adaptation, as well as downlink beamforming through channel reciprocity. From the sensing perspective, however, the CSI is itself a snapshot of the environment. The distance to a reflector appears in the delay components of the channel, and its motion appears as symbol-to-symbol phase variation, i.e., Doppler.

The sensing block converts the CSI into an RDMap through the preprocessing described in Section 2.2 and detects events from it with the AI model described in Section 2.3. The RDMap is a two-dimensional map of Doppler velocity and bistatic range (in Figure 2, the horizontal axis is Doppler velocity and the vertical axis is bistatic range); the intensity of each cell indicates the energy reflected from objects at that range and

velocity. Because the preprocessing aligns all paths to the LoS path, range is measured relative to the LoS path, which is why the range axis in Figure 2

2.2 Signal Preprocessing

Figure 2 shows the preprocessing chain with measured results at each stage, taken from an offline study using over-the-air measurements. A commercial Galaxy S25+ smartphone transmitted a 5G New Radio (NR) uplink reference signal over 100 MHz at 3.85 GHz to a 64TR Massive MIMO Unit (MMU) installed on a building at the Seoul R&D Campus; the UE was placed at the entrance of another building across the street, with pedestrians moving in between. This offline study uses the full received data, while the real-time system described in Section 2.4 uses a reduced subset.

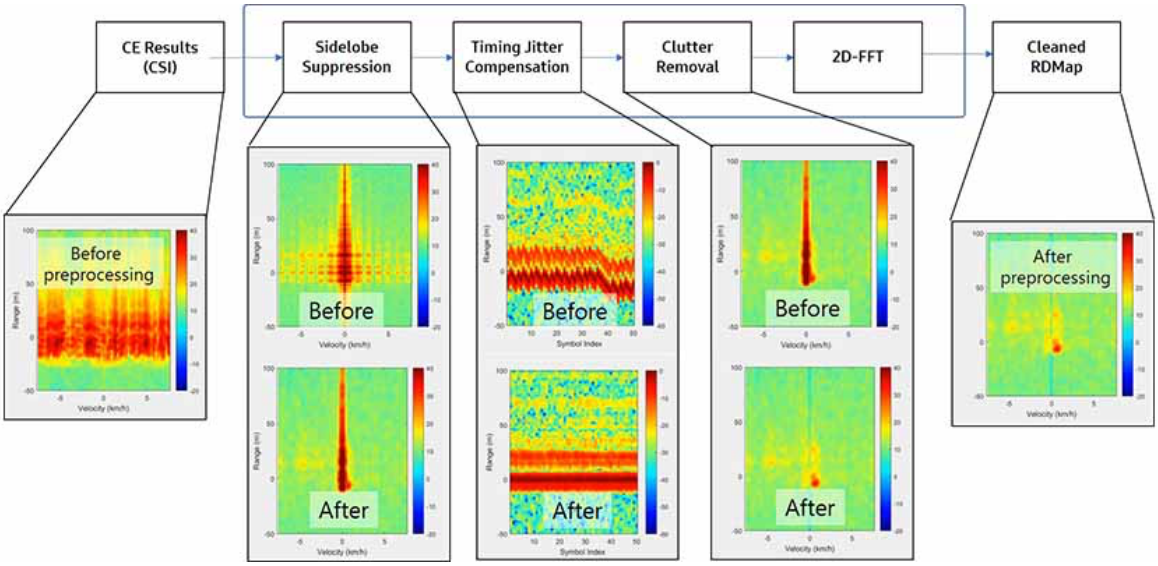


Figure 2. Signal preprocessing blocks and measured results at each stage.

The input is the frequency-domain CSI, a matrix of subcarriers by reference-signal symbols. An IFFT along the subcarriers and an FFT along the symbols turn it into an RDMap. Applying these transforms directly to raw CSI yields the leftmost map: energy is smeared across the velocity axis by timing errors, static clutter, and LoS sidelobes, and no target can be discerned.

Sidelobe Suppression

Finite bandwidth and a finite observation time produce sinc-shaped sidelobes around strong paths such as the LoS path; these sidelobes can be tens of dB stronger than human reflections. Windowing along frequency and time removes the resulting stripes at the cost of a slightly wider main lobe.

Timing Jitter Compensation

Although the UE is synchronized to the BS for communication, residual symbol-to-symbol timing fluctuations from its own oscillator and timing adjustments make the LoS range drift over the processing interval [2], [6], which smears energy along the Doppler axis and makes static objects appear to move. With the LoS path as a reference, the per-symbol offset is estimated and removed. Without this step, meaningful Doppler analysis is not possible in bistatic ISAC.

Clutter Removal

Walls, floors, and the LoS path form a strong ridge at zero Doppler. Removing this static component reveals the weak reflections beneath it; the strength of this removal must be tuned so that the subtle motion of a nearly stationary person is not removed with it.
The cleaned RDMap on the right reveals targets that were completely hidden in the raw map and is passed to the AI-based event detection block in Figure 1.

AI-based Event Detection

At first glance, an RDMap looks like an image, but it differs from natural images in ways that matter for model design. Learning-based methods have shown strong potential across wireless problems [1], yet in sensing their benefit depends on whether the inductive biases of the model match the physics of the radio channel.

What makes RDMap data different. Doppler carries absolute meaning: zero Doppler separates static from moving reflectors, and its sign distinguishes motion toward or away from the link, so a horizontal flip, a routine image augmentation, changes the label itself. Multipath creates weaker ghost copies of a target at longer bistatic ranges. Moving structures in the environment, such as ordinary and revolving doors, produce Doppler of their own that can resemble human motion. Finally, noise is always present: targets are weak, the noise floor varies with link conditions, and residual clutter remains after preprocessing.

Target events. The model distinguishes four states: enter (a person entering), exit (a person leaving), presence (someone nearby without passing through), and clear (no one present). The Doppler sign indicates whether the bistatic path is shortening or lengthening, but how this maps to entering or leaving depends on where the entrance lies relative to the UE-BS geometry. In practice, apart from the clear case, the four states are difficult to tell apart by eye in individual RDMaps. This is especially true at the entrance with the automatic revolving door, which starts rotating

whenever someone approaches: as Figure 3 shows, presence, enter, and exit all produce energy spread over a similar span of range and Doppler, and the differences lie in subtle patterns that evolve over consecutive frames. This is precisely where a learned model is needed.

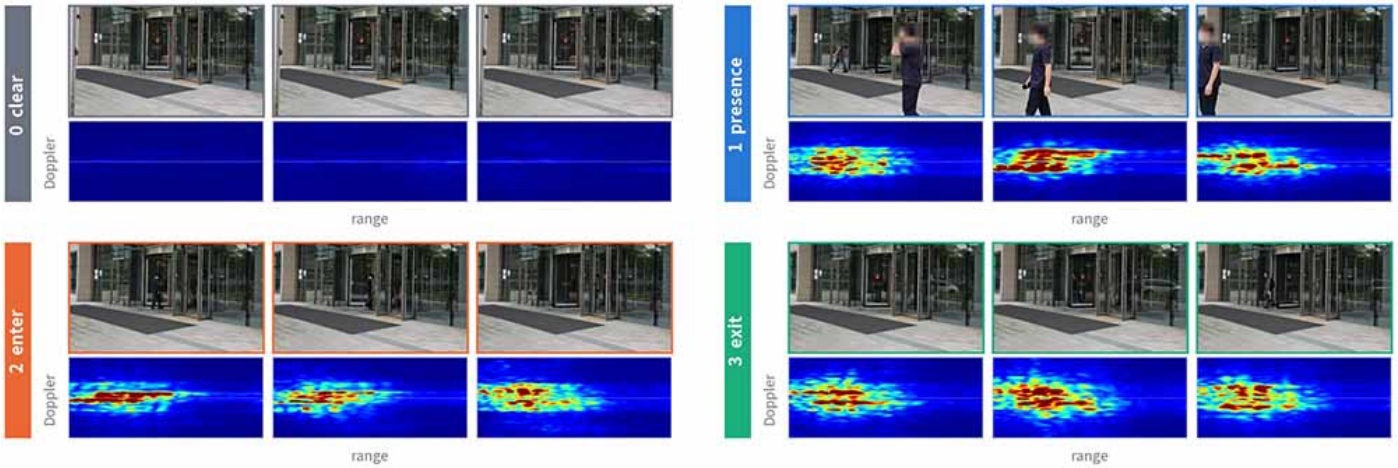


Figure 3. Camera frames (used only for labeling) and the corresponding RDMaps for the four states at the entrance with the automatic revolving door (range on the horizontal axis, Doppler on the vertical axis). Apart from the clear state, the RDMaps are difficult to distinguish by eye.

Design principles. Rather than describing a specific architecture, we highlight the principles that mattered most in practice. Decisions are made over short sequences of consecutive RDMaps rather than single frames. The meaning of the Doppler axis is preserved, and data augmentation is limited to transformations a real channel could produce. Because each site has its own static structures, moving doors, and residual clutter, the model learns the background of the deployment environment. For evaluation, we use held-out sessions, because random frame-level splits overestimate accuracy when consecutive frames are strongly correlated. Finally, the model is kept lightweight enough for real-time inference at the 200 ms update rate.

2.4 Integration into the RAN

A sensing chain that works offline must still fit within the radio resources, data paths, and computing resources of a commercial RAN, none of which were dimensioned with sensing in mind.

Reference signal period. We evaluated both dense and sparse sounding of the reference signal and confirmed that the difference in detection performance is marginal, so the sensing period can be relaxed without sacrificing detection.

Data volume. Full-band CSI is large. For a 100 MHz carrier at 30 kHz subcarrier spacing (3,276 subcarriers), 64 receive antennas, and 16-bit I/Q samples, it amounts to hundreds of Mbit/s for a single UE at typical sounding periods. While the offline study in Figure 2 uses the full data, the real-time system reduces it in two dimensions before it leaves the distributed unit (DU). In frequency, we keep only a small subset of subcarriers spread across the band; range resolution is unchanged, because it is set by the total bandwidth, and although the maximum unambiguous range shrinks, it remains far beyond the sensing area of about 100 m. In space, we use a subset of the receive antennas. Together, these steps reduce the data by more than an order of magnitude compared with full-band CSI from all antennas.

Stable real-time output. The model outputs a state every 200 ms, and individual outputs can fluctuate from one update to the next. Rather than reporting raw outputs, we apply post-processing with event-specific decision thresholds, which keeps the reported state stable while still following people as they pass, at the cost of a short delay.

System integration. Channel estimation runs on the MMU as it does for communication. The DU forwards the reduced CSI to an external server equipped with a GPU, which runs preprocessing, AI inference, and post-processing in real time. The system runs on a commercial vRAN, in which the DU is implemented as software on a general-purpose server. This allowed us to add the sensing data path through software updates alone, so that new versions could be deployed and tested on the live network quickly and easily. If the vRAN server itself is equipped with a GPU, the sensing function can share it with the RAN software [8], and the entire chain can run within a single platform. We instead built the pipeline to operate with an external GPU server so that sensing can also be added to existing deployments without GPU acceleration. The radio hardware is commercial and unmodified, and sensing needs no dedicated sensing signal or additional radio unit. Running inference on a separate server also keeps the sensing computational load off the RAN, so that it does not compete with communication processing, and the data reduction above keeps the forwarded data rate modest. Figure 4 shows the resulting configuration. Only event-level results are exposed to applications.

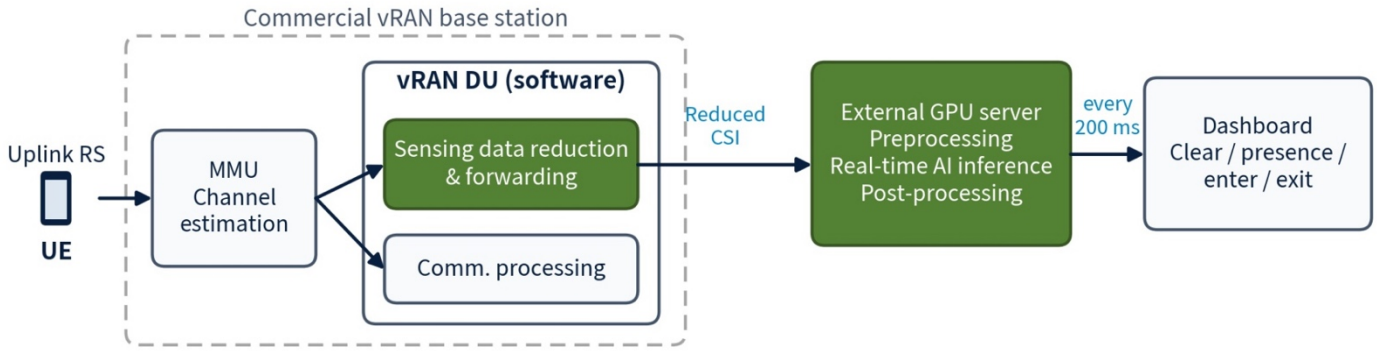


Figure 4. Real-time system integration: the DU forwards reduced CSI to an external GPU server for preprocessing, inference, and post-processing.

3. Implementation Results

3.1 Experimental Environment

Experiments were conducted on our campus with a commercial smartphone and a commercial vRAN-based base station. Table 1 summarizes the experimental environment.

Table 1. Experimental environment

Item	Specification
Base station	Commercial 64TR MMU and vRAN-based DU, 3.85 GHz, 100 MHz bandwidth, 30 kHz subcarrier spacing
UE	Galaxy S25+, stationary, indoors near a building entrance
Sensing signal	Uplink reference signal
CSI used	Full for the preprocessing study (Figure 2); a subset of subcarriers and antennas for event detection
Real-time processing	Reduced data forwarded from the DU to an external GPU server; inference every 200 ms with post-processing
Test sites	Two campus entrances (one with two ordinary doors; one with an ordinary door and a revolving door)
Data collection	Various times of day and numbers of people
Output	Four states: clear, presence, enter, exit

3.2 Entry and Exit Detection

A commercial smartphone (Galaxy S25+) was placed indoors in a fixed position near a building entrance, and a base station outside the building received its uplink reference signals. The smartphone stood in for a dedicated fixed sensing device or customer premises equipment (CPE), which would take this role in deployment without additional hardware. People walking through the entrance crossed the propagation paths between the UE and the BS; because the UE did not move, the geometry was stable, and changes in the RDMap could be attributed to the environment. Figure 5 shows one of the test sites: the UE was deployed indoors behind an entrance with a revolving door, and the MMU was mounted on the rooftop of a building across the street.

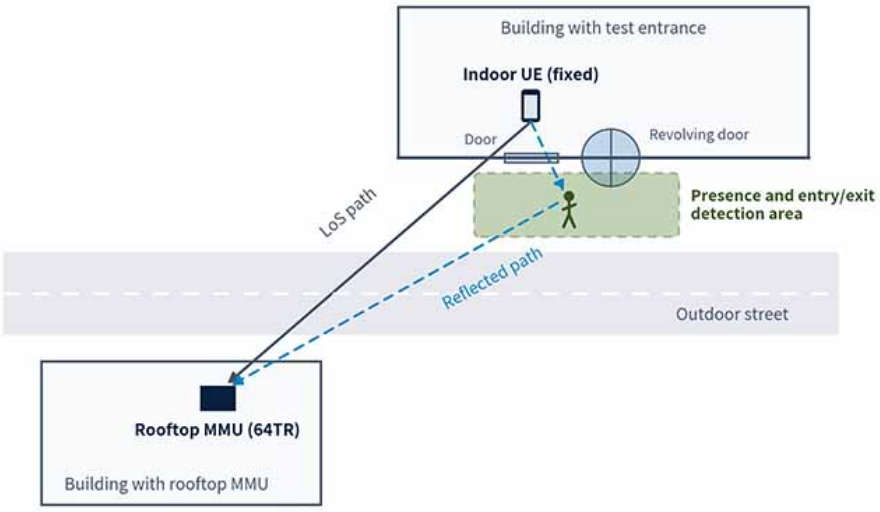


Figure 5. Test site layout: the UE deployed indoors at a building entrance with a revolving door, and the MMU mounted on the rooftop of a building across the street. Both the LoS path and paths reflected by people near the entrance reach the MMU.

We collected data at two entrances on our campus, covering different times of day and numbers of people. One entrance has two ordinary doors; the other combines an ordinary door with an automatic revolving door, which starts rotating whenever a person approaches and adds a Doppler signature of its own. Across both sites, the system consistently distinguished the four states, including at the entrance with the automatic revolving door (Figure 3), where the model had to separate a person walking through from a door that moves whenever someone is nearby. Such entry and exit events support occupancy management, building security, and energy control.

We also ran the same chain in real time with the live uplink of the UE (Section 2.4), displaying the detected state on a dashboard every 200 ms, so that the result could be seen changing as people walked through the entrance. This real-time system was further demonstrated to several mobile network operators; unlike the data collection on our campus, in which recorded data were evaluated offline, the demonstrations showed live inference results on site. Figure 6 shows the dashboard at the test entrance: as a person leaves the building, the detected state changes to exit, shown alongside the latest RDMap.

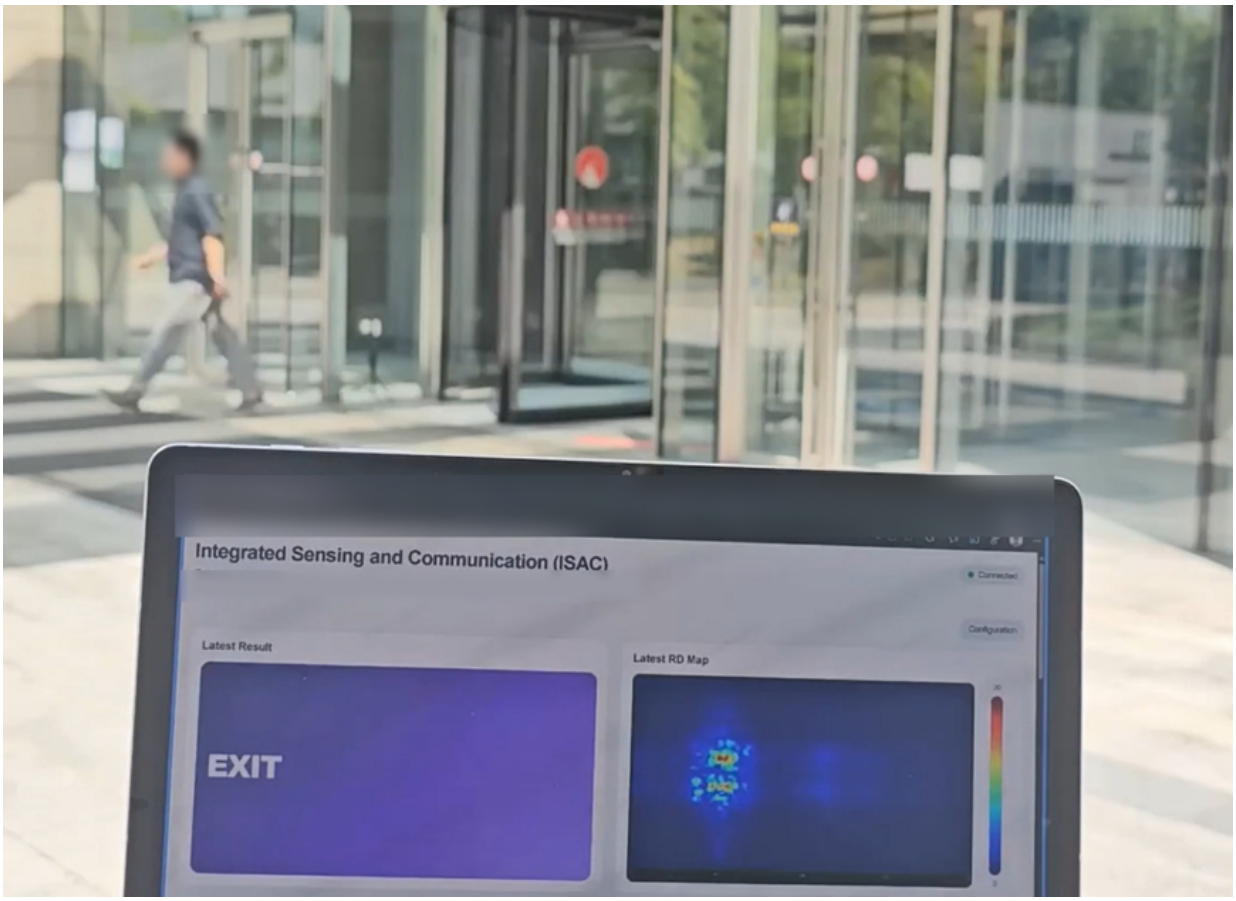


Figure 6. Real-time operation at the test entrance: the dashboard shows the detected state (exit) and the latest RDMap as a person leaves the building.

4. Conclusion

This post has presented a practical path to ISAC on a commercial vRAN-based 5G network. By reusing the uplink CSI that the receiver already produces, our system detects presence and entry/exit at a building entrance using only a base station and an ordinary UE, without additional radio hardware.

Key Contributions:

· Sensing on a commercial vRAN: sensing is added to a commercial vRAN-based network through software updates, reusing uplink reference signals on unmodified radio hardware, with inference on an external GPU server.
· A practical preprocessing chain for bistatic CSI: sidelobe suppression, timing jitter compensation, and clutter removal, validated on over-the-air measurements, turn unusable raw CSI into clean RDMaps.
· Validated in practice: experiments at two campus entrances, including one with a revolving door, real-time operation on a commercial vRAN, and demonstrations to several mobile network operators.

Future work will focus on two directions. The first is on-site learning. Because each site has its own static structures, moving doors, and residual clutter (Section 2.3), we aim to let the model adapt to a new location using data collected on site, so that the system can be deployed at new buildings without retraining from scratch. The second concerns operation at scale. When multiple UEs sense multiple locations within a cell, reference signal resources must be shared among the sensing links and with communication. This calls for a sensing-aware scheduler that decides which UEs to sound, how often, and when, for example by assigning sparse sounding to idle locations and dense sounding to active ones, building on our observation that detection performance is maintained even with sparse sounding. Privacy remains a design principle throughout: the system reports only clear, presence, enter, and exit states, contains no identifying information, and processes data at the edge.

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